

HR ANALYTICS REPORT

Employee Data Analysis (2006–2019)

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Role: Data Analyst

Tool Used: Microsoft Excel (Power Query, PivotTables, PivotCharts, Slicers, Dashboard)

Reporting Period: 2006–2019

Executive Summary

This report presents the findings from an HR analytics project conducted on employee data spanning **2006 to 2019**. The objective was to clean and transform the dataset, develop an interactive Excel dashboard, and analyze key workforce metrics to support strategic human resource decision-making.

The analysis focused on employee turnover, workforce demographics, tenure, recruitment effectiveness, compensation, employee engagement, job satisfaction, and performance. Interactive PivotTables, PivotCharts, KPI cards, and slicers were developed to allow users to dynamically explore the data by department, performance rating, and gender.

The findings reveal an overall employee turnover rate of **33.4%**, with the **Production** department experiencing the highest turnover. Employee terminations peaked in **2015**, while **Indeed** emerged as the most effective recruitment source, producing both the highest number of hires and the largest number of high-performing employees. Employees rated **Exceeds Expectations** demonstrated the longest average tenure, whereas employees in the **IT/IS** department tended to leave the organization sooner than those in Production. The report concludes with practical recommendations aimed at improving employee retention, recruitment strategies, and workforce engagement.

Project Objectives

The objectives of this project were to:

- Clean and prepare the HR dataset for analysis.
- Develop an interactive HR analytics dashboard in Microsoft Excel.

- Analyze employee turnover and workforce trends.
- Evaluate recruitment source effectiveness.
- Assess employee performance, engagement, and satisfaction.
- Identify potential salary disparities.
- Generate actionable insights to support HR decision-making.

Methodology

The project was completed entirely in Microsoft Excel.

The following activities were performed:

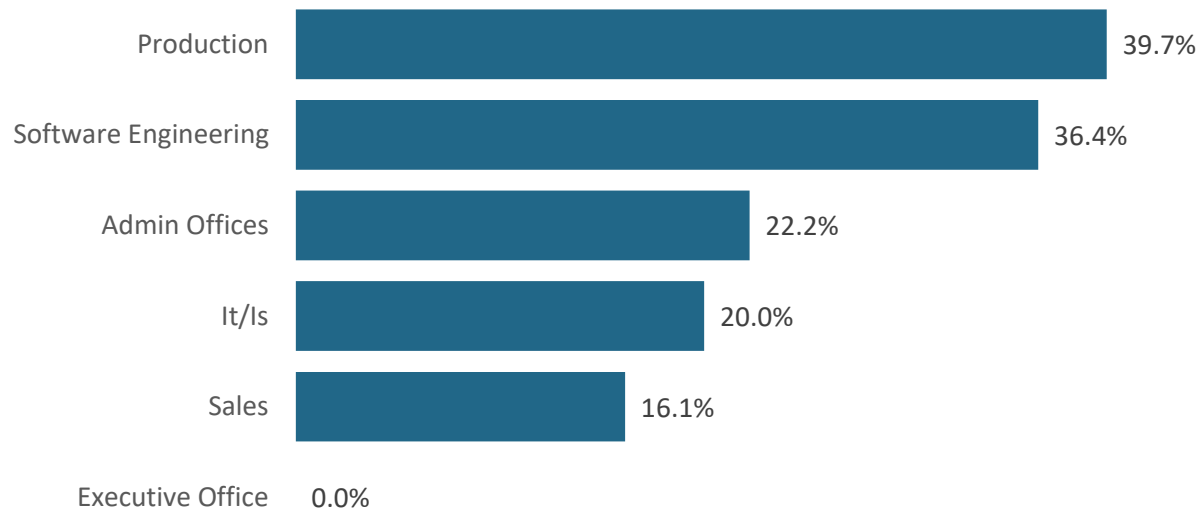
- Data cleaning and transformation using Power Query.
- Correction of employee age and tenure calculations.
- Standardization of text fields and employment status.
- Creation of calculated fields including turnover rate, active employees, tenure, and age groups.
- Development of PivotTables and PivotCharts.
- Creation of KPI cards and an interactive dashboard using slicers.
- Analysis of workforce trends to answer key business questions.

Business Findings

Business Question 1

What is the overall turnover rate? Which department has the highest turnover? Which has the lowest?

Employee Turnover Rate by Department



Findings

The organization recorded an overall employee turnover rate of **33.4%** during the reporting period.

Among all departments:

- **Production** recorded the highest turnover rate (**39.7%**).
- **Software Engineering** followed with **36.4%**.
- **Executive Office** recorded **0% turnover**, making it the department with the lowest turnover.

Insight

The significantly higher turnover within the Production department suggests persistent employee retention challenges that may be associated with workload, working conditions, leadership, or career development opportunities.

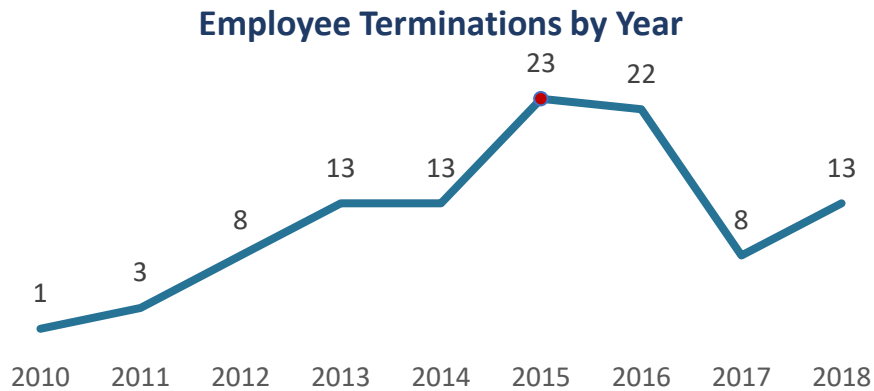
Recommendation

HR should conduct a detailed retention analysis within the Production department by reviewing employee engagement, workload distribution, compensation, career progression, and supervisory practices.

Business Question 2

Which year between 2010 and 2018 had the most terminations?

Findings



Employee terminations increased steadily from **2010** before reaching a peak in **2015**, when **23 employees** left the organization. Terminations remained relatively high in **2016** with **22 exits**, before declining in subsequent years.

Insight

The spike in employee exits during 2015–2016 suggests a period of increased workforce instability that may have resulted from organizational restructuring, leadership changes, or external labour market conditions.

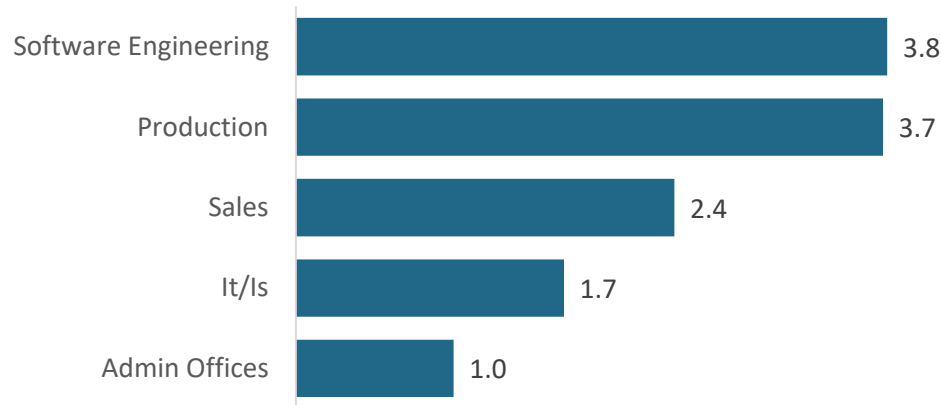
Recommendation

HR should review organizational initiatives, management practices, and employee feedback from the 2015–2016 period to identify the factors that contributed to increased turnover and implement preventive measures where appropriate.

Business Question 3

What is the average tenure of employees who leave? Do Production employees leave sooner than IT/IS employees?

Average Tenure of Terminated Employees by Department



Findings

The average tenure of employees who left the organization was **3.41 years**.

Employees in the **Production** department remained with the organization for an average of **3.72 years**, while employees in the **IT/IS** department stayed for an average of **1.70 years** before leaving.

Insight

Although Production records the highest turnover rate, employees in the IT/IS department leave much earlier. This suggests that IT/IS experiences greater challenges with early employee retention.

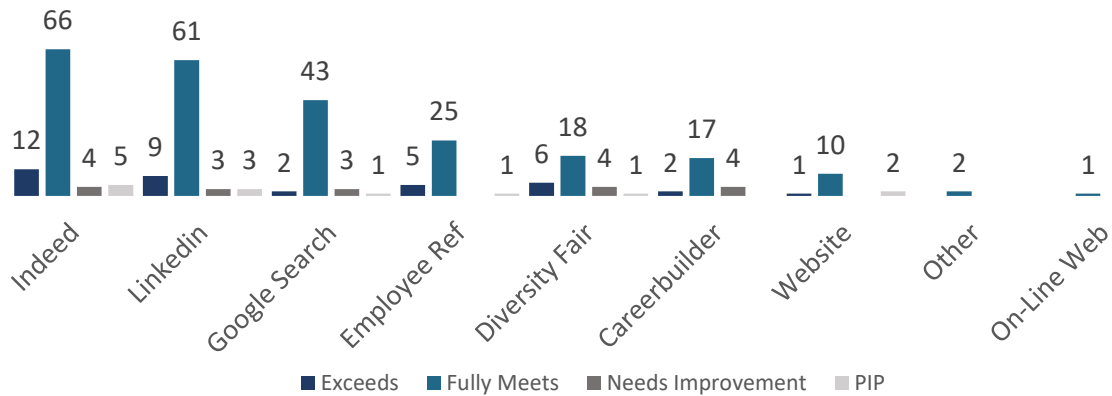
Recommendation

HR should investigate the causes of early employee exits within the IT/IS department through exit interviews, employee engagement assessments, career development initiatives, and compensation reviews.

Business Question 4

Which recruitment source produces the most hires? Which source produces the highest-performing employees?

Recruitment Sources: Hires and Performance Ratings



Findings

Indeed generated the highest number of employee hires (**87**) and also produced the highest number of employees rated **Exceeds Expectations (12)**.

LinkedIn ranked second with **76 hires** and **9 employees** rated as Exceeds Expectations.

Insight

Indeed consistently attracts both a high volume of candidates and a strong proportion of high-performing employees, making it the most effective recruitment source within the organization.

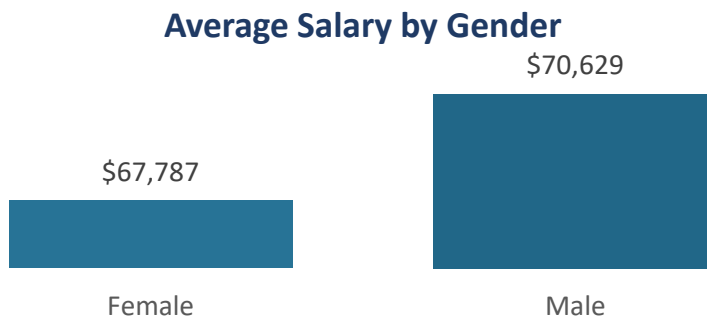
Recommendation

HR should continue prioritizing recruitment through Indeed and LinkedIn while evaluating the return on investment of lower-performing recruitment channels.

Business Question 5

Is there a gender pay gap? What is the average salary for males versus females?

Findings



The average salary for male employees was **\$70,629**, while female employees earned an average salary of **\$67,787**.

The average salary difference between male and female employees was approximately **\$2,842**.

Insight

Although a salary difference exists, further analysis is required to determine whether it is influenced by job role, department, tenure, or experience rather than gender alone.

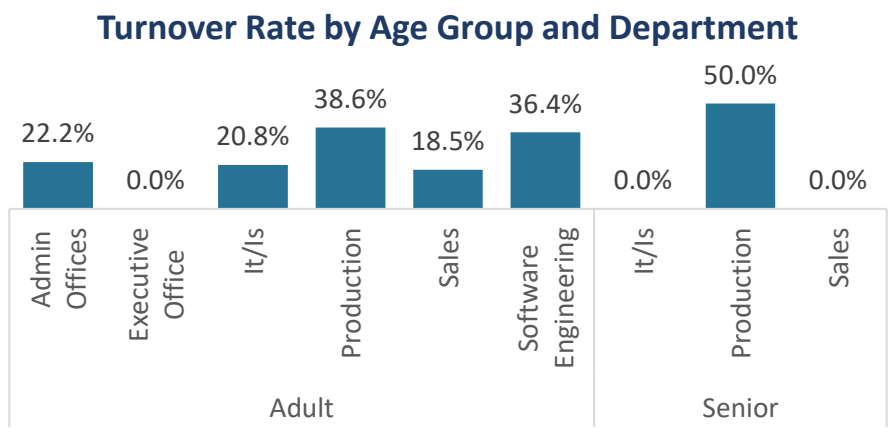
Recommendation

HR should periodically conduct salary equity reviews to ensure compensation practices remain fair, transparent, and aligned with organizational policies.

Business Question 6

Which age group has the highest turnover rate?

Findings



The **Senior** age group recorded the highest turnover rate at **38.5%**, compared to approximately **33.0%** among the Adult age group.

Within the Senior age group, the Production department experienced the highest turnover.

Insight

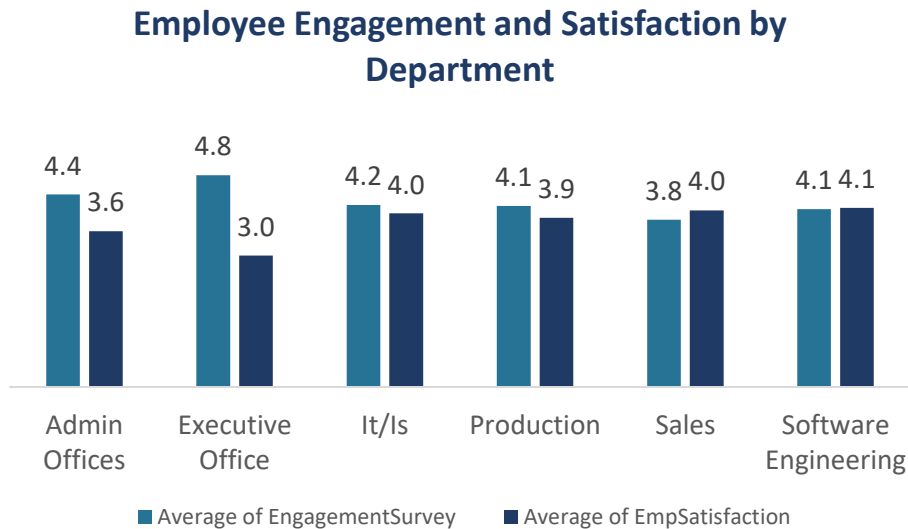
The higher turnover among senior employees may indicate increased retirements, succession planning challenges, or retention issues affecting experienced personnel.

Recommendation

HR should strengthen succession planning initiatives and implement retention strategies aimed at preserving experienced employees and facilitating effective knowledge transfer.

Business Question 7

Which department has the highest average engagement score? Which has the highest average satisfaction?



Findings

The **Executive Office** recorded the highest average employee engagement score (**4.8**).

The **Software Engineering** department recorded the highest average employee satisfaction score (**4.1**).

Insight

The results suggest that employee engagement and employee satisfaction measure different aspects of the employee experience. High engagement does not necessarily correspond to the highest satisfaction.

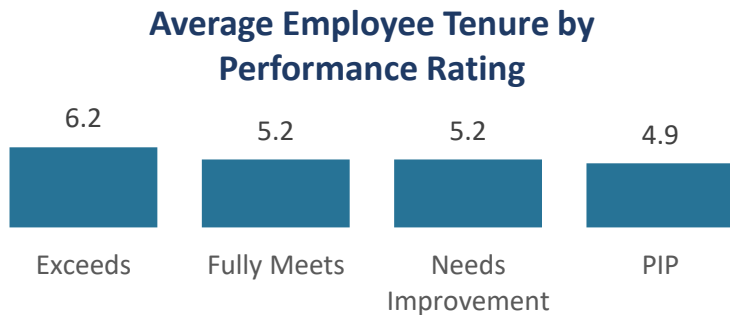
Recommendation

HR should evaluate the practices contributing to high engagement within the Executive Office and identify opportunities to replicate successful employee experience initiatives across other departments.

Business Question 8

Do employees who exceed expectations stay longer than those who need improvement?

Findings



Employees rated **Exceeds Expectations** remained with the organization for an average of **6.2 years**.

Employees rated **Needs Improvement** remained for an average of **5.2 years**, while employees on **Performance Improvement Plans (PIP)** had the shortest average tenure at **4.9 years**.

Insight

Higher-performing employees generally remain with the organization longer, indicating a positive relationship between employee performance and employee retention.

Recommendation

HR should continue investing in employee development, recognition programmes, mentoring, and career advancement opportunities to encourage high performance and improve long-term employee retention.

Overall Recommendations

Based on the analysis, the following strategic actions are recommended:

- Strengthen employee retention initiatives within the Production department.
- Investigate the causes of early employee exits in the IT/IS department.
- Continue investing in high-performing recruitment sources such as Indeed and LinkedIn.
- Conduct regular salary equity assessments to promote fair compensation practices.
- Expand employee engagement and satisfaction initiatives across departments.
- Enhance career development and recognition programmes to retain high-performing employees.
- Improve succession planning for experienced employees approaching retirement.
- Continue monitoring workforce metrics using the interactive HR Analytics dashboard to support evidence-based decision-making.

Conclusion

The HR Analytics Dashboard successfully transformed raw employee data into meaningful business insights that support strategic workforce planning and informed decision-making.

The analysis identified key trends in employee turnover, recruitment effectiveness, workforce demographics, compensation, engagement, satisfaction, and performance. By integrating data cleaning, interactive dashboards, PivotTables, and visual analytics within Microsoft Excel, the project demonstrates how HR data can be leveraged to improve organizational performance.

The interactive dashboard provides HR leaders with a practical decision-support tool for monitoring workforce metrics, identifying emerging trends, and implementing targeted interventions to improve employee retention and overall organizational effectiveness.